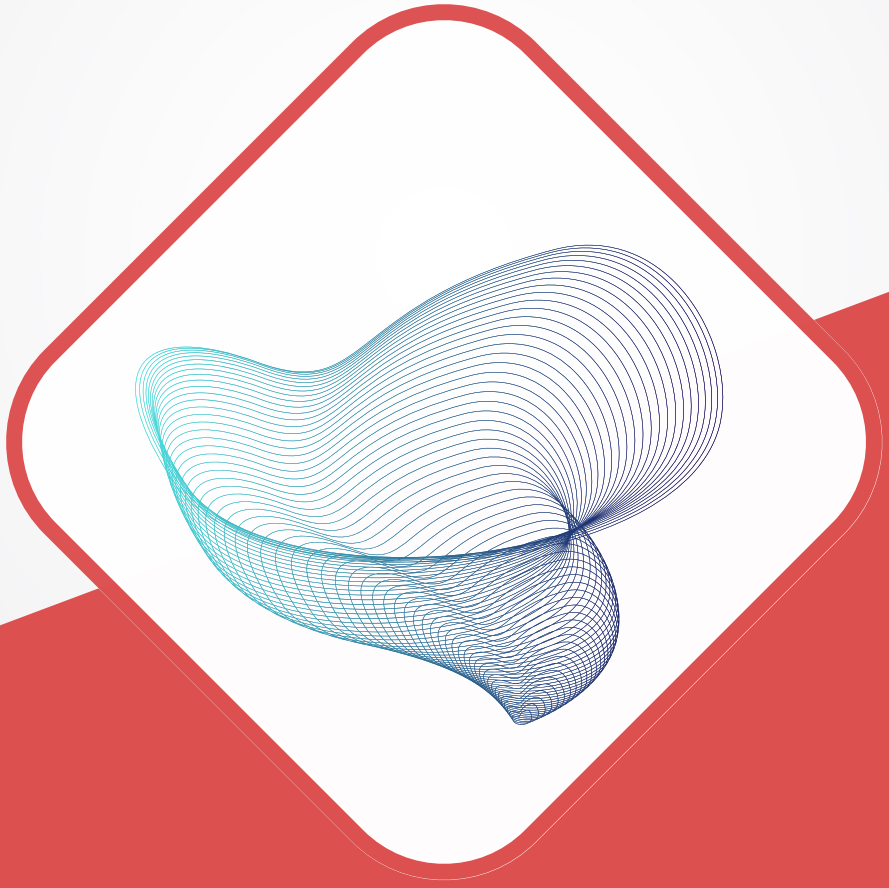
ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)

path to success

JEE (MAIN+ADVANCED)

Mathematics

ENTHUSIAST COURSE



STUDY MATERIAL

Vector-3D

English Medium

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3D-COORDINATE GEOMETRY

POINT

1. INTRODUCTION :

In earlier classes we have learnt about points, lines, circles and conic section in two dimensional geometry. In two dimensions a point represented by an ordered pair (x, y) (where x & y are both real numbers)

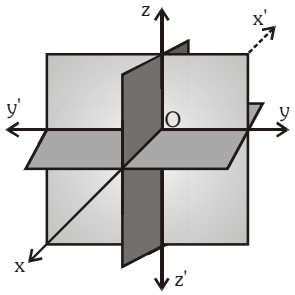
In space, each body has length, breadth and height i.e. each body exist in three dimensional space. Therefore three independent quantities are essential to represent any point in space. Three axes are required to represent these three quantities.

2. RECTANGULAR CO-ORDINATE SYSTEM :

In cartesian system of three lines which are mutually perpendicular, such a system is called rectangular cartesian co-ordinate system.

Co-ordinate axes and co-ordinate planes :

When three mutually perpendicular planes intersect at a point, then mutually perpendicular lines are obtained and these lines also pass through that point. If we assume the point of intersection as origin, then the three planes are known as co-ordinate planes and the three lines are known as co-ordinate axes.

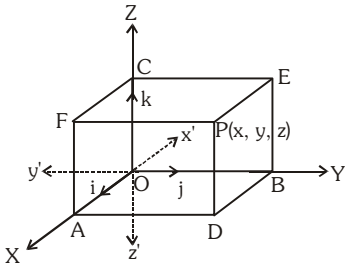


Octants :

Every plane bisects the space. Hence three co-ordinate plane divide the space in eight parts. These parts are known as octants.

3. COORDINATES OF A POINT IN SPACE :

Let O be a fixed point, known as origin and let OX, OY and OZ be three mutually perpendicular lines, taken as x-axis, y-axis and z-axis respectively, in such a way that they form a right handed system.



The planes XOY, YOZ and ZOX are known as xy-plane, yz-plane and zx-plane respectively.

Let P be a point in space and distances of P from yz, zx and xy planes be x, y, z respectively (with proper signs) then we say that coordinates of P are (x, y, z). Also OA = |x|, OB = |y|, OC = |z|

4. DISTANCE FORMULA :

The distance between two points A (x_1, y_1, z_1) and B (x_2, y_2, z_2) is given by

$$AB = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2]}$$

(a) Distance from Origin :

Let O be the origin and P (x, y, z) be any point, then $OP = \sqrt{(x^2 + y^2 + z^2)}$

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(b) Distance of a point from coordinate axes :

Let P(x, y, z) be any point in the space. Let PA, PB and PC be the perpendiculars drawn from P to the axes OX, OY and OZ respectively. Then

$PA = \sqrt{(y^2+z^2)}$; $PB = \sqrt{(z^2+x^2)}$; $PC = \sqrt{(x^2+y^2)}$

Illustration 1 : Prove by using distance formula that the points P (1, 2, 3), Q (–1, –1, –1) and R(3, 5, 7) are collinear.

Solution : We have $PQ = \sqrt{(-1-1)^2+(-1-2)^2+(-1-3)^2} = \sqrt{4+9+16} = \sqrt{29}$

$$QR = \sqrt{(3+1)^2+(5+1)^2+(7+1)^2} = \sqrt{16+36+64} = \sqrt{116} = 2\sqrt{29}$$

and $PR = \sqrt{(3-1)^2+(5-2)^2+(7-3)^2} = \sqrt{4+9+16} = \sqrt{29}$

Since QR = PQ + PR. Therefore the given points are collinear. **Ans.**

Illustration 2 : Find the locus of a point the sum of whose distances from (1, 0, 0) and (–1, 0, 0) is equal to 10.

Solution : Let the points A(1,0,0), B (–1,0,0) and P(x,y,z)

Given : PA + PB =10

$$\sqrt{(x-1)^2+(y-0)^2+(z-0)^2} + \sqrt{(x+1)^2+(y-0)^2+(z-0)^2} = 10$$

$$\Rightarrow \quad \sqrt{(x-1)^2+y^2+z^2} = 10 - \sqrt{(x+1)^2+y^2+z^2}$$

Squaring both sides, we get ;

$$\Rightarrow \quad (x-1)^2+y^2+z^2 = 100 + (x+1)^2+y^2+z^2 - 20\sqrt{(x+1)^2+y^2+z^2}$$

$$\Rightarrow \quad -4x-100 = -20\sqrt{(x+1)^2+y^2+z^2} \quad \Rightarrow \quad x+25 = 5\sqrt{(x+1)^2+y^2+z^2}$$

Again squaring both sides we get $x^2 + 50x + 625 = 25\ \{(x^2+2x+1) + y^2 + z^2\}$

$$\Rightarrow \quad 24x^2 + 25y^2 + 25z^2 - 600 = 0$$

i.e. required equation of locus **Ans.**

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5. SECTION FORMULAE :

Let $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ be two points and let R (x, y, z) divide PQ in the ratio $m_1 : m_2$. Then co-ordinates of $R(x, y, z) = \left(\frac{m_1x_2+m_2x_1}{m_1+m_2}, \frac{m_1y_2+m_2y_1}{m_1+m_2}, \frac{m_1z_2+m_2z_1}{m_1+m_2}\right)$

If (m_1/m_2) is positive, R divides PQ internally and if (m_1/m_2) is negative, then externally.

Mid-Point : Mid point of PQ is given by $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}, \frac{z_1+z_2}{2}\right)$

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Illustration 3 : Find the ratio in which the plane x – 2y + 3z = 17 divides the line joining the points (–2, 4, 7) and (3, –5, 8).

Solution : Let the required ratio be k : 1

The co-ordinates of the point which divides the join of (–2, 4, 7) and (3, –5, 8) in the ratio k : 1 are $\left(\frac{3k-2}{k+1}, \frac{-5k+4}{k+1}, \frac{8k+7}{k+1}\right)$

Since this point lies on the plane x – 2y + 3z – 17 = 0

$\therefore \quad \left(\frac{3k-2}{k+1}\right)-2\left(\frac{-5k+4}{k+1}\right)+3\left(\frac{8k+7}{k+1}\right)-17=0$

$\Rightarrow$ (3k – 2) –2 (–5k + 4) +3 (8k + 7) = 17 k + 17

$\Rightarrow$ 3k + 10k + 24k – 17k = 17 + 2 + 8 – 21

$\Rightarrow \quad 37k - 17k = 6 \quad \Rightarrow \quad 20k = 6 ; \quad k = \frac{6}{20} = \frac{3}{10}$

Hence the required ratio = k : 1 $= \frac{3}{10} : 1 = 3 : 10$ **Ans.**

6. CENTROID OF A TRIANGLE :

Let $A(x_1, y_1, z_1)$, $B(x_2, y_2, z_2)$, $C(x_3, y_3, z_3)$ be the vertices of a triangle ABC. Then its centroid G is given by $G = \left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}, \frac{z_1+z_2+z_3}{3}\right)$

Illustration 4 : If the centre of a tetrahedron OABC where A, B, C, are given by (a, 2, 3), (1, b, 2) and (2, 1, c) respectively is (1, 2, – 1), then distance of P (a, b, c) from origin is -

(A) $\sqrt{107}$ (B) $\sqrt{14}$ (C) $\sqrt{107}/14$ (D) none of these

Solution : Centre is $\left(\frac{1}{4}\Sigma x, \frac{1}{4}\Sigma y, \frac{1}{4}\Sigma z\right) = (1, 2, -1)$

$\Rightarrow \quad \frac{a+1+2+0}{4}=1, \frac{2+b+1+0}{4}=2, \frac{3+2+c+0}{4}=-1 \quad \Rightarrow \quad a = 1, b = 5, c = -9$

$\therefore \quad OP = \sqrt{a^2+b^2+c^2} = \sqrt{107}$ **Ans. (A)**

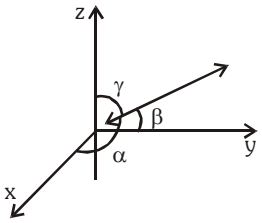
Do yourself 1:

1. Find the distance between the points P(3, 4, 5) and Q(–1, 2, –3).
2. Show that the points A(0, 7, 10), B(–1, 6, 6) and C(–4, 9, 6) are vertices of an isosceles right angled triangle.
3. Find the locus of a point such that the difference of the square of its distance from the points A(3, 4, 5) and B(–1, 3, –7) is equal to $2k^2$.
4. Find the co-ordinates of points which trisects the line joining the points A(–3, 2, 4) and B(0, 4, 7)
5. Find the ratio in which the planes (a) xy (b) yz divide the line joining the points P(–2, 4, 7) and Q(3, –5, 8).

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7. DIRECTION COSINES OF LINE :

If α, β, γ be the angles made by a line with x-axis, y-axis & z-axis respectively then $\cos\alpha$, $\cos\beta$ & $\cos\gamma$ are called direction cosines of a line, denoted by ℓ, m & n respectively.



Note :

(i) If line makes angles α, β, γ with x, y & z axis respectively then $\pi - \alpha$, $\pi - \beta$ & $\pi - \gamma$ is another set of angle that line makes with principle axes. Hence if ℓ, m & n are direction cosines of line then $-\ell$, $-$m & $-$n are also direction cosines of the same line.

(ii) Since parallel lines have same direction. So, in case of lines, which do not pass through the origin. We can draw a parallel line passing through the origin and direction cosines of that line can be found.

Important points :

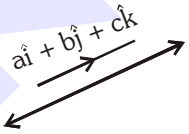
(i) Direction cosines of a line :

Take a vector $\vec{A} = a\hat{i} + b\hat{j} + c\hat{k}$ parallel to a line whose D.C's are to be found out.

$\vec{A}.\hat{i} = a$

$|\vec{A}|\cos\alpha = a$

$\cos\alpha = \frac{a}{|\vec{A}|}$ similarly, $\cos\beta = \frac{b}{|\vec{A}|}$; $\cos\gamma = \frac{c}{|\vec{A}|}$



$\Rightarrow \quad \cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1 \quad \Rightarrow \quad \ell^2 + m^2 + n^2 = 1$

(ii) Direction cosine of axes :

Since the positive x-axes makes angle 0°, 90°, 90° with axes of x, y and z respectively,

$\therefore$ D.C.'s of x axes are 1, 0, 0.

D.C.'s of y-axis are 0, 1, 0

D.C.'s of z-axis are 0, 0, 1

8. DIRECTION RATIOS :

Any three numbers a, b, c proportional to direction cosines ℓ, m, n are called direction ratios of the line. i.e. $\frac{\ell}{a} = \frac{m}{b} = \frac{n}{c}$

There can be infinitely many sets of direction ratios for a given line.

Direction ratios and Direction cosines of the line joining two points :

Let $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$ be two points, then d.r.'s of AB are $x_2 - x_1$, $y_2 - y_1$, $z_2 - z_1$ and the d.c.'s of AB are $\frac{1}{r}(x_2 - x_1)$, $\frac{1}{r}(y_2 - y_1)$, $\frac{1}{r}(z_2 - z_1)$ where $r = \sqrt{[\Sigma(x_2 - x_1)^2]}$

9. RELATION BETWEEN D.C'S & D.R'S :

$$\frac{\ell}{a} = \frac{m}{b} = \frac{n}{c}$$

$$\therefore \quad \frac{\ell^2}{a^2} = \frac{m^2}{b^2} = \frac{n^2}{c^2} = \frac{\ell^2 + m^2 + n^2}{a^2 + b^2 + c^2}$$

$$\therefore \quad \ell = \frac{\pm a}{\sqrt{a^2 + b^2 + c^2}} \; ; \; m = \frac{\pm b}{\sqrt{a^2 + b^2 + c^2}} \; ; \; n = \frac{\pm c}{\sqrt{a^2 + b^2 + c^2}}$$

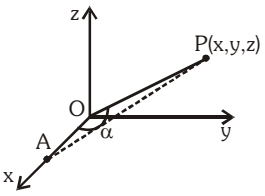
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Important point :

Direction cosines of a line have two sets but direction ratios of a line have infinite possible sets.

10. PROJECTIONS :

(a) Length of Projection of line segment OP on co-ordinate axes :



Let line segment make angle α with x-axis

Thus, the projections of line segment OP on axes are the absolute values of the co-ordinates of P. i.e.

Length of Projection of OP on x-axis = |x|

Length of Projection of OP on y-axis = |y|

Length of Projection of OP on z-axis = |z|

Now, in ΔOAP, angle A is a right angle and OA = x

$$OP = \sqrt{x^2 + y^2 + z^2}$$

$$\therefore \quad \cos\alpha = \frac{x}{\sqrt{x^2 + y^2 + z^2}} = \frac{x}{|OP|}$$

if |OP| = r, then $x = |OP|\cos\alpha = \ell r$

Similarly $y = |OP|\cos\beta = mr$, $z = nr$, where ℓ, m, n are DC's of line

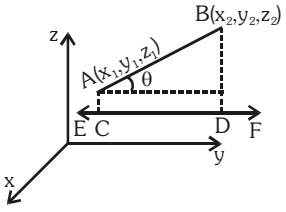
(b) Length of Projection of a line segment AB on coordinate axes :

Projection of the point $A(x_1, y_1, z_1)$ on x-axis is $E(x_1, 0, 0)$. Projection of point $B(x_2, y_2, z_2)$ on x-axis is $F(x_2, 0, 0)$.

Hence Length of projection of AB on x-axis is $EF = |x_2 - x_1|$.

Similarly, Length of projection of AB on y and z-axis are $|y_2 - y_1|$, $|z_2 - z_1|$ respectively.

(c) Length of **Projection of line segment AB on a line having direction cosines ℓ, m, n :**



Let $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$.

Now Length of projection of AB on EF = CD = AB cosθ

$$= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2} \times \frac{\left|(x_2 - x_1)\ell + (y_2 - y_1)m + (z_2 - z_1)n\right|}{\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}}$$

$$= \left|(x_2 - x_1)\ell + (y_2 - y_1)m + (z_2 - z_1)n\right|$$

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Illustration 5 : A line OP makes with the x-axis an angle of measure 120° and with y-axis an angle of measure 60°. Find the angle made by the line with the z-axis.

Solution : $\alpha = 120°$ and $\beta = 60°$

$\therefore \cos\alpha = \cos 120° = -\frac{1}{2}$ and $\cos\beta = \cos 60° = \frac{1}{2}$ but $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

$\therefore \quad \left(\frac{-1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 + \cos^2\gamma = 1$

$\Rightarrow \quad \cos^2\gamma = 1 - \frac{1}{4} - \frac{1}{4} = \frac{1}{2} \quad \Rightarrow \cos\gamma = \pm\frac{1}{\sqrt{2}}$

$\therefore \quad \gamma = 45° \quad$ or $\quad 135°$ **Ans.**

Illustration 6 : Find the length of projection of the line segment joining the points (–1, 0, 3) and (2, 5, 1) on the line whose direction ratios are 6, 2, 3.

Solution : The direction cosines ℓ, m, n of the line are given by $\frac{\ell}{6} = \frac{m}{2} = \frac{n}{3} = \frac{\sqrt{\ell^2+m^2+n^2}}{\sqrt{6^2+2^2+3^2}} = \frac{1}{\sqrt{49}} = \frac{1}{7}$

$\therefore \quad \ell = \frac{6}{7}, m = \frac{2}{7}, n = \frac{3}{7}$

The required length of projection is given by

$= |\ell(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1)| = \left|\frac{6}{7}[2-(-1)] + \frac{2}{7}(5-0) + \frac{3}{7}(1-3)\right|$

$= \left|\frac{6}{7}\times 3 + \frac{2}{7}\times 5 + \frac{3}{7}\times -2\right| = \left|\frac{18}{7} + \frac{10}{7} - \frac{6}{7}\right| = \left|\frac{18+10-6}{7}\right| = \frac{22}{7}$ **Ans.**

Do yourself - 2 :

1. Find the length of projections of the line segment joining the origin O to the point P(3, 2, –5) on the axes.
2. Find the length of projections of the line joining the points P(3, 2, 5) and Q(0, –2, 8) on the axes.
3. Find the direction ratios & direction cosines of the line joining the points O(0, 0, 0) and P(2, 3, 4).

11. ANGLE BETWEEN TWO LINES :

Let θ be the angle between the lines with d.c.'s ℓ_1, m_1, n_1 and ℓ_2, m_2, n_2 then $\cos\theta = \ell_1\ell_2 + m_1m_2 + n_1n_2$.

If a_1, b_1, c_1 and a_2, b_2, c_2 be D.R.'s of two lines then angle θ between them is given by

$$\cos\theta = \frac{(a_1a_2 + b_1b_2 + c_1c_2)}{\sqrt{(a_1^2+b_1^2+c_1^2)}\sqrt{(a_2^2+b_2^2+c_2^2)}}$$

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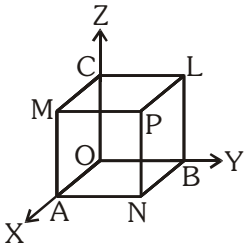
Illustration 7 : If a line makes angles $\alpha, \beta, \gamma, \delta$ with four diagonals of a cube, then $\cos^2\alpha + \cos^2\beta + \cos^2\gamma + \cos^2\delta$ equals -

(A) 3 (B) 4 (C) 4/3 (D) 3/4

Solution : Let OA, OB, OC be coterminous edges of a cube and OA = OB = OC = a, then co-ordinates of its vertices are O(0, 0, 0), A(a, 0, 0), B(0, a, 0), C(0, 0, a), L(0, a, a), M(a, 0, a), N(a, a, 0) and P (a, a, a)

Direction ratio of diagonal AL, BM, CN and OP are

$$\left(-\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}\right),\left(\frac{1}{\sqrt{3}},-\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}\right),\left(\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}},-\frac{1}{\sqrt{3}}\right),\left(\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}\right)$$



Let ℓ, m, n be the direction cosines of the given line, then

$$\cos\alpha = \ell\left(-\frac{1}{\sqrt{3}}\right)+m\left(\frac{1}{\sqrt{3}}\right)+n\left(\frac{1}{\sqrt{3}}\right)=\frac{-\ell+m+n}{\sqrt{3}}$$

Similarly $\cos\beta = \frac{\ell-m+n}{\sqrt{3}}$, $\cos\gamma = \frac{\ell+m-n}{\sqrt{3}}$ and $\cos\delta = \frac{\ell+m+n}{\sqrt{3}}$

$\therefore$ $\cos^2\alpha + \cos^2\beta + \cos^2\gamma + \cos^2\delta = \frac{4}{3}$ **Ans. (C)**

Illustration 8 : (a) Find the acute angle between two lines whose direction ratios are 2, 3, 6 and 1, 2, 2 respectively.

(b) Find the measure of the angle between the lines whose direction ratios are 1, –2, 7 and 3, –2, –1.

Solution : (a) $a_1 = 2, b_1 = 3, c_1 = 6; a_2 = 1, b_2 = 2, c_2 = 2.$

If θ be the angle between two lines whose d.r's are given, then

$$\cos\theta = \frac{a_1a_2+b_1b_2+c_1c_2}{\sqrt{a_1^2+b_1^2+c_1^2}\sqrt{a_2^2+b_2^2+c_2^2}} = \frac{2\times1+3\times2+6\times2}{\sqrt{2^2+3^2+6^2}\sqrt{1^2+2^2+2^2}} = \frac{2+6+12}{7\times3}=\frac{20}{21}$$

$\therefore$ $\theta = \cos^{-1}\left(\frac{20}{21}\right)$

(b) $\sqrt{1^2+(-2)^2+7^2} = \sqrt{54}$

$\sqrt{3^2+(-2)^2+(-1)^2} = \sqrt{14}$

$\therefore$ The actual direction cosines of the lines are

$\frac{1}{\sqrt{54}}, \frac{-2}{\sqrt{54}}, \frac{7}{\sqrt{54}}$ and $\frac{3}{\sqrt{14}}, \frac{-2}{\sqrt{14}}, \frac{-1}{\sqrt{14}}$

If θ is the angle between the lines, then

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$$\cos\theta = \left(\frac{1}{\sqrt{54}}\right)\left(\frac{3}{\sqrt{14}}\right)+\left(\frac{-2}{\sqrt{54}}\right)\left(\frac{-2}{\sqrt{14}}\right)+\left(\frac{7}{\sqrt{54}}\right)\left(\frac{-1}{\sqrt{14}}\right)$$

$$= \frac{3+4-7}{\sqrt{54}.\sqrt{14}} \qquad = 0 \quad \Rightarrow \quad \theta = 90° \qquad \textbf{Ans.}$$

12. PERPENDICULAR AND PARALLEL LINES :

Let the two lines have their d.c.'s given by ℓ_1, m_1, n_1 and ℓ_2, m_2, n_2 respectively then they are perpendicular if $\theta = 90°$ i.e. $\cos\theta = 0$, i.e. $\ell_1\ell_2 + m_1m_2 + n_1n_2 = 0$.

Also the two lines are parallel if $\theta = 0$ i.e. $\sin\theta = 0$, i.e. $\frac{\ell_1}{\ell_2} = \frac{m_1}{m_2} = \frac{n_1}{n_2}$

Note: If instead of d.c.'s, d.r.'s a_1, b_1, c_1 and a_2, b_2, c_2 are given, then the lines are perpendicular if $a_1a_2 + b_1b_2 + c_1c_2 = 0$ and parallel if $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$.

Illustration 9 : If the lines whose direction cosines satisfies the equations $a\ell + bm + cn = 0$ and $fmn + gn\ell + h\ell m = 0$ are perpendicular, then $\frac{f}{a}+\frac{g}{b}+\frac{h}{c}$ equals -

(A) 0 (B) –1 (C) 1 (D) none of these

Solution : Eliminating n between the given relations, we find that $(fm + g\ell)\left(\frac{-a\ell - bm}{c}\right) + h\ell m = 0$

or $$ag\left(\frac{\ell}{m}\right)^2 + (af + bg - ch)\left(\frac{\ell}{m}\right) + bf = 0 \qquad .....(i)$$

Let $\frac{\ell_1}{m_1}$ and $\frac{\ell_2}{m_2}$, are roots of (i), then $\frac{\ell_1}{m_1}\cdot\frac{\ell_2}{m_2} = \frac{bf}{ag}$

$$\Rightarrow \quad \frac{\ell_1\ell_2}{f/a} = \frac{m_1m_2}{g/b} \qquad .....(ii)$$

Similarly $$\frac{m_1m_2}{g/b} = \frac{n_1n_2}{h/c} \qquad .....(iii)$$

From (ii) and (iii), we get $$\frac{\ell_1\ell_2}{f/a} = \frac{m_1m_2}{g/b} = \frac{n_1n_2}{h/c} = \lambda$$

$$\Rightarrow \quad \ell_1\ell_2 = \lambda.f/a \ ; \ m_1m_2 = \lambda.g/b \ ; \ n_1n_2 = \lambda.h/c$$

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$\Rightarrow \quad \ell_1\ell_2 + m_1m_2 + n_1n_2 = \lambda\left(\frac{f}{a}+\frac{g}{b}+\frac{h}{c}\right)$

$\Rightarrow \quad \frac{f}{a}+\frac{g}{b}+\frac{h}{c} = 0 \qquad \{\because\ \ell_1\ell_2 + m_1m_2 + n_1n_2 = 0\}$ **Ans. (A)**

Do yourself - 3 :

1. Find the angle between the lines whose direction ratios are 1, –2, 1 and 4, 3, 2.

2. If a line makes α, β and γ angle with axes, then prove that $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$.

3. Find the direction cosines of the line which is perpendicular to the lines with direction cosines proportional to (1, –2, –2) & (0, 2, 1).

4. Find the angle between the two straight lines whose direction cosines ℓ, m, n are given by $2\ell + 2m - n = 0$ and $mn + n\ell + \ell m = 0$.

5. The acute angle between two lines such that the direction cosines ℓ, m, n of each of them satisfy the equations $\ell + m + n = 0$ and $\ell^2 + m^2 - n^2 = 0$ is :-

(A) 30° (B) 45° (C) 60° (D) 15°

6. If two lines L_1 and L_2 in space, are definedby

$L_1 = \left\{x = \sqrt{\lambda}\ y + \left(\sqrt{\lambda} - 1\right)\right.$

$\left.z = \left(\sqrt{\lambda} - 1\right) y + \sqrt{\lambda}\right\}$ and

$L_2 = \left\{x = \sqrt{\mu}\ y + \left(1 - \sqrt{\mu}\right)\right.$

$\left.z = \left(1 - \sqrt{\mu}\right) y + \sqrt{\mu}\right\}$, then L_1 is perpendicular to L_2, for all non-negative reals λ and μ, such that :

(A) $\lambda = \mu$ (B) $\lambda \neq \mu$ (C) $\sqrt{\lambda} + \sqrt{\mu} = 1$ (D) $\lambda + \mu = 0$

PLANE

13. DEFINITION :

A plane is a surface such that a line segment joining any two points on the surface lies wholly on it.

14. EQUATIONS OF A PLANE :

The equation of every plane is of the first degree i.e. of the form $ax + by + cz + d = 0$, in which a, b, c are constants, not all zero simultaneously.

(a) Equation of plane passing through a fixed point :

Vector form : If $\vec{a}$ is the position vector of a point on the plane and $\vec{n}$ is a vector normal to the plane then its vectorial equation is given by $(\vec{r} - \vec{a}).\vec{n} = 0 \Rightarrow \mathbf{\vec{r}.\vec{n} = d}$, where $d = \vec{a}.\vec{n}$ = constant.

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Cartesian form : If $\vec{a}(x_1, y_1, z_1)$ and $\vec{n} = a\hat{i} + b\hat{j} + c\hat{k}$, then cartesian equation of plane will be

$a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$

(b) Plane Parallel to the Coordinate Planes :

(i) Equation of yz plane is x = 0.

(ii) Equation of zx plane is y = 0.

(iii) Equation of xy plane is z = 0.

(iv) Equation of the plane parallel to xy plane at a distance c is z = c or z = –c.

(v) Equation of the plane parallel to yz plane at a distance c is x = c or x = –c

(vi) Equation of the plane parallel to zx plane at a distance c is y = c or y = –c.

(c) Equations of Planes Parallel to the Axes :

If a = 0, the plane is parallel to x-axis i.e. equation of the plane parallel to x-axis is **by + cz + d = 0**.

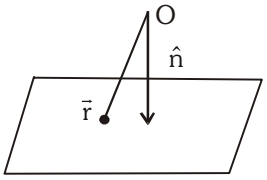
Similarly, equations of planes parallel to y-axis and parallel to z-axis are **ax + cz + d = 0** and **ax + by + d = 0**, respectively.

(d) Equation of a Plane in Intercept Form :

Equation of the plane which cuts off intercepts a, b, c from the axes x, y, z respectively is

$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$.

(e) Equation of a Plane in Normal Form :



Vector form : If $\hat{n}$ is a unit vector normal to the plane from the origin and d be the perpendicular distance of plane from origin then its vector equation is $\vec{r} \cdot \hat{n} = d$.

Cartesian form : If the length of the perpendicular distance of the plane from the origin is p and direction cosines of this perpendicular are (ℓ, m, n), then the equation of the plane is **$\ell x + my + nz = p$**.

(f) Equation of a Plane through three points :

Vector form : If A, B, C are three points having P.V.'s $\vec{a}$, $\vec{b}$, $\vec{c}$ respectively, then vector equation of the plane is $[\vec{r}\ \vec{a}\ \vec{b}] + [\vec{r}\ \vec{b}\ \vec{c}] + [\vec{r}\ \vec{c}\ \vec{a}] = [\vec{a}\ \vec{b}\ \vec{c}]$.

Cartesian form : The equation of the plane through three non-collinear points (x_1, y_1, z_1), (x_2, y_2, z_2) and (x_3, y_3, z_3) is $\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ x_3 - x_1 & y_3 - y_1 & z_3 - z_1 \end{vmatrix} = 0$

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Illustration 10 : Find the equation of the plane through the points A(2, 2, –1), B(3, 4, 2) and C(7, 0, 6).

Solution : The general equation of a plane passing through (2, 2, –1) is

$$a(x-2)+b(y-2)+c(z+1)=0 \qquad \text{.....(i)}$$

It will pass through B (3, 4, 2) and C (7, 0, 6) if

$$a(3-2)+b(4-2)+c(2+1)=0 \quad \text{or} \quad a+2b+3c=0 \qquad \text{.....(ii)}$$

and $a(7-2)+b(0-2)+c(6+1)=0$ or $5a-2b+7c=0$ .....(iii)

Solving (ii) and (iii) by cross-multiplication, we have

$$\frac{a}{14+6}=\frac{b}{15-7}=\frac{c}{-2-10} \text{ or } \frac{a}{5}=\frac{b}{2}=\frac{c}{-3}=\lambda \qquad \text{(say)}$$

$\Rightarrow$ $a = 5\lambda$, $b = 2\lambda$ and $c = -3\lambda$

Substituting the values of a, b and c in (i), we get

$$5\lambda(x-2)+2\lambda(y-2)-3\lambda(z+1)=0$$

or $5(x-2)+2(y-2)-3(z+1)=0$

$\Rightarrow$ $5x + 2y - 3z = 17$, which is the required equation of the plane **Ans.**

Illustration 11 : A plane meets the co-ordinates axes in A,B,C such that the centroid of the Δ ABC is the point (p,q,r) show that the equation of the plane is $\frac{x}{p}+\frac{y}{q}+\frac{z}{r}=3$

Solution : Let the required equation of plane be :

$$\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1 \qquad \text{.....(i)}$$

Then, the co-ordinates of A, B and C are A(a, 0, 0), B(0, b, 0), C(0, 0, c) respectively

So the centroid of the triangle ABC is $\left(\frac{a}{3},\frac{b}{3},\frac{c}{3}\right)$

But the co-ordinate of the centroid are (p,q,r)

$$\frac{a}{3}=p,\ \frac{b}{3}=q,\ \frac{c}{3}=r$$

Putting the values of a, b and c in (i), we get the required plane as $\frac{x}{3p}+\frac{y}{3q}+\frac{z}{3r}=1$

$\Rightarrow$ $\frac{x}{p}+\frac{y}{q}+\frac{z}{r}=3$ **Ans.**

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Do yourself - 4 :

1. Equation of a plane is $3x + 4y + 5z = 7$.

(a) Find the direction cosines of its normal

(b) Find the points where it intersects the axes.

(c) Find its intercept form.

(d) Find its equation in normal form (in cartesian as well as in vector form)

2. Find the equation of the plane passing through the points (2, 3, 1), (3, 0, 2) and (–1, 2, 3).

3. A vector $\vec{n}$ is inclined to x-axis at 45°, to y–axis at 60° and at an acute angle to z-axis. If $\vec{n}$ is a normal to a plane passing through the point ($\sqrt{2}$, –1, 1), then the equation of the plane is :

(A) $\sqrt{2}\,x - y - z = 2$ (B) $\sqrt{2}\,x + y + z = 2$

(C) $3\sqrt{2}\,x - 4y - 3z = 7$ (D) $4\sqrt{2}\,x + 7y + z = 2$

4. Consider three vectors $\vec{p} = \hat{i} + \hat{j} + \hat{k}$, $\vec{q} = 2\hat{i} + 4\hat{j} - \hat{k}$ and $\vec{r} = \hat{i} + \hat{j} + 3\hat{k}$. If $\vec{p}$, $\vec{q}$ and $\vec{r}$ denotes the position vector of three non-collinear points then the equation of the plane containing these points is

(A) $2x - 3y + 1 = 0$ (B) $x - 3y + 2z = 0$

(C) $3x - y + z - 3 = 0$ (D) $3x - y - 2 = 0$

15. ANGLE BETWEEN TWO PLANES :

Vector form : If $\vec{r}\,.\,\vec{n}_1 = d_1$ and $\vec{r}\,.\,\vec{n}_2 = d_2$ be two planes, then angle between these planes is the angle between their normals

$$\cos\theta = \frac{\vec{n}_1 \,.\, \vec{n}_2}{|\vec{n}_1|\,|\vec{n}_2|}$$

$\therefore$ Planes are perpendicular if $\vec{n}_1 \,.\, \vec{n}_2 = 0$ and they are parallel if $\vec{n}_1 = \lambda\,\vec{n}_2$.

Cartesian form : Consider two planes $ax + by + cz + d = 0$ and $a'x + b'y + c'z + d' = 0$. Angle between these planes is the angle between their normals.

$$\cos\theta = \frac{aa' + bb' + cc'}{\sqrt{a^2 + b^2 + c^2}\,\sqrt{a'^2 + b'^2 + c'^2}}$$

$\therefore$ Planes are perpendicular if $aa' + bb' + cc' = 0$ and they are parallel if $\frac{a}{a'} = \frac{b}{b'} = \frac{c}{c'}$.

Planes parallel to a given Plane :

Equation of a plane parallel to the plane $ax + by + cz + d = 0$ is $ax + by + cz + d' = 0$. d' is to be found by other given condition.

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Illustration 12 : Find the angle between the planes $x + y + 2z = 9$ and $2x - y + z = 15$

Solution : We know that the angle between the planes $a_1x + b_1y + c_1z + d_1 = 0$ and $a_2x + b_2y + c_2z + d_2 = 0$ is given by $\cos\theta = \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}}$

Therefore, angle between $x + y + 2z = 9$ and $2x - y + z = 15$ is given by

$$\cos\theta = \frac{(1)(2)+(1)(-1)+(2)(1)}{\sqrt{1^2+1^2+2^2}\sqrt{2^2+(-1)^2+1^2}} = \frac{1}{2} \quad \Rightarrow \theta = \frac{\pi}{3}$$ **Ans.**

Illustration 13 : Find the equation of the plane through the point (1, 4, –2) and parallel to the plane $-2x + y - 3z = 7$.

Solution : Let the equation of a plane parallel to the plane $-2x + y - 3z = 7$ be $-2x + y - 3z + k = 0$

This passes through (1, 4, –2), therefore $(-2)(1) + 4 - 3(-2) + k = 0$

$\Rightarrow \quad -2 + 4 + 6 + k = 0 \quad \Rightarrow \quad k = -8$

Putting $k = -8$ in (i), we obtain $-2x + y - 3z - 8 = 0$ or $-2x + y - 3z = 8$ **Ans.**

This is the equation of the required plane.

Do yourself - 5 :

1. Prove that the planes $3x - 2y + z + 17 = 0$ and $4x + 3y - 6z - 25 = 0$ are perpendicular.

2. Find the angle between the planes $3x + 4y + z + 7 = 0$ and $-x + y - 2z = 5$

16. A PLANE THROUGH THE LINE OF INTERSECTION OF TWO GIVEN PLANES :

Consider two planes $u \equiv ax + by + cz + d = 0$ and $v \equiv a'x + b'y + c'z + d' = 0$.

The equation $u + \lambda v = 0$, λ a real parameter, represents the plane passing through the line of intersection of given planes and if planes are parallel, this represents a plane parallel to them.

Illustration 14 : Find the equation of plane containing the line of intersection of the plane $x + y + z - 6 = 0$ and $2x + 3y + 4z + 5 = 0$ and passing through (1,1,1).

Solution : The equation of the plane through the line of intersection of the given planes is,

$$(x + y + z - 6) + \lambda(2x + 3y + 4z + 5) = 0 \quad \text{.....(i)}$$

If it is passes through (1,1,1)

$\Rightarrow \quad (1 + 1 + 1 - 6) + \lambda(2 + 3 + 4 + 5) = 0 \quad \Rightarrow \quad \lambda = \frac{3}{14}$

Putting $\lambda = 3/14$ in (i); we get $(x + y + z - 6) + \frac{3}{14}(2x + 3y + 4z + 5) = 0$

$\Rightarrow \quad 20x + 23y + 26z - 69 = 0$ **Ans.**

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17. PERPENDICULAR DISTANCE OF A POINT FROM THE PLANE :

Vector form : If $\vec{r}.\vec{n} = d$ be the plane, then perpendicular distance p, of the point A($\vec{a}$)

$$p = \frac{|\vec{a}.\vec{n} - d|}{|\vec{n}|}$$

Distance between two parallel planes $\vec{r}.\vec{n} = d_1$ & $\vec{r}.\vec{n} = d_2$ is $\left|\frac{d_1 - d_2}{|\vec{n}|}\right|$.

Cartesian form : Perpendicular distance p, of the point $A(x_1, y_1, z_1)$ from the plane $ax + by + cz + d = 0$ is given by $p = \frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{(a^2 + b^2 + c^2)}}$

Distance between two parallel planes $ax + by + cz + d_1 = 0$ & $ax + by + cz + d_2 = 0$ is $\left|\frac{d_1 - d_2}{\sqrt{a^2 + b^2 + c^2}}\right|$

Illustration 15 : Find the perpendicular distance of the point (2, 1, 0) from the plane $2x + y + 2z + 5 = 0$

Solution : We know that the perpendicular distance of the point (x_1, y_1, z_1) from the plane $ax + by + cz + d = 0$ is $\frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}}$

so required distance $= \frac{|2\times2 + 1\times1 + 2\times0 + 5|}{\sqrt{2^2 + 1^2 + 2^2}} = \frac{10}{3}$ **Ans.**

Illustration 16 : Find the distance between the parallel planes $2x - y + 2z + 3 = 0$ and $4x - 2y + 4z + 5 = 0$.

Solution : Let $P(x_1, y_1, z_1)$ be any point on $2x - y + 2z + 3 = 0$, then $2x_1 - y_1 + 2z_1 + 3 = 0$

The length of the perpendicular from $P(x_1, y_1, z_1)$ to $4x - 2y + 4z + 5 = 0$ is

$$\left|\frac{4x_1 - 2y_1 + 4z_1 + 5}{\sqrt{4^2 + (-2)^2 + 4^2}}\right| = \left|\frac{2(2x_1 - y_1 + 2z_1) + 5}{\sqrt{36}}\right| = \frac{|2(-3) + 5|}{6} = \frac{1}{6} \text{ [using (i)]}$$

Therefore, the distance between the two given parallel planes is $\frac{1}{6}$ **Ans.**

Do yourself - 6 :

1. Find the perpendicular distance of the point P(1, 2, 3) from the plane $2x + y + z + 1 = 0$.

2. Find the equation of the plane passing through the line of intersection of the planes $x + y + z = 5$ and $2x + 3y + z + 5 = 0$ and passing through the point (0, 0, 0).

3. The equation of a plane passing through the line of intersection of the planes $x + 2y + 3z = 2$ and $x - y + z = 3$ and at a distance $\frac{2}{\sqrt{3}}$ from the point (3, 1, –1) is

(A) $5x - 11y + z = 17$ (B) $\sqrt{2}x + y = 3\sqrt{2} - 1$

(C) $x + y + z = \sqrt{3}$ (D) $x - \sqrt{2}y = 1 - \sqrt{2}$

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18. BISECTORS OF ANGLES BETWEEN TWO PLANES :

Let the equations of the two planes be $ax + by + cz + d = 0$ and $a_1x + b_1y + c_1z + d_1 = 0$.
Then equations of bisectors of angles between them are given by

$$\frac{ax+by+cz+d}{\sqrt{(a^2+b^2+c^2)}} = \pm\frac{a_1x+b_1y+c_1z+d_1}{\sqrt{(a_1^2+b_1^2+c_1^2)}}$$

(a) Equation of bisector of the angle containing origin : First make both constant terms positive. Then positive sign give the bisector of the angle which contains the origin.

(b) Bisector of acute/obtuse angle : First making both constant terms positive,

$aa_1 + bb_1 + cc_1 > 0 \quad \Rightarrow$ origin lies in obtuse angle

$aa_1 + bb_1 + cc_1 < 0 \quad \Rightarrow$ origin lies in acute angle

---

Illustration 17 : Find the equation of the bisector planes of the angles between the planes $2x - y + 2z + 3 = 0$ and $3x - 2y + 6z + 8 = 0$ and specify the plane which bisects the acute angle and the plane which bisects the obtuse angle.

Solution : The two given planes are $2x - y + 2z + 3 = 0$ and $3x - 2y + 6z + 8 = 0$

where $d_1, d_2 > 0$

and $a_1a_2 + b_1b_2 + c_1c_2 = 6 + 2 + 12 > 0$

$$\therefore \quad \frac{a_1x+b_1y+c_1z+d_1}{\sqrt{a_1^2+b_1^2+c_1^2}} = -\frac{a_2x+b_2y+c_2z+d_2}{\sqrt{a_2^2+b_2^2+c_2^2}} \text{ (acute angle bisector)}$$

$$\text{and} \quad \frac{a_1x+b_1y+c_1z+d}{\sqrt{a_1^2+b_1^2+c_1^2}} = \frac{a_2x+b_2y+c_2z+d_2}{\sqrt{a_2^2+b_2^2+c_2^2}} \text{ (obtuse angle bisector)}$$

$$\text{i.e.,} \quad \frac{2x-y+2z+3}{\sqrt{4+1+4}} = \pm\frac{3x-2y+6z+8}{\sqrt{9+4+36}}$$

$\Rightarrow \quad (14x - 7y + 14z + 21) = \pm (9x - 6y + 18z + 24)$

Taking positive sign on the right hand side,

we get $5x - y - 4z - 3 = 0$ (obtuse angle bisector)

and taking negative sign on the right hand side,

we get $23x - 13y + 32z + 45 = 0$ (acute angle bisector) **Ans.**

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19. POSITION OF TWO POINTS W.R.T. A PLANE :

Two points $P(x_1, y_1, z_1)$ & $Q(x_2, y_2, z_2)$ are on the same or opposite sides of a plane $ax + by + cz + d = 0$ according to $ax_1 + by_1 + cz_1 + d$ & $ax_2 + by_2 + cz_2 + d$ are of same or opposite signs. The plane divides the line joining the points P & Q externally or internally according to P and Q lying on same or opposite sides of the plane.

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Do yourself - 7 :

1. Find the position of the point P(2, –2, 1), Q(3, 0, 1) and R(–12, 1, 8) w.r.t. the plane $2x - 3y + 4z - 7 = 0$.

2. Two given planes are $-2x + y - 2z + 5 = 0$ and $6x - 2y + 3z - 7 = 0$. Find

(a) equation of plane bisecting the angle between the planes.

(b) equation of a plane parallel to the plane bisecting the angle between both the two planes and passing through the point (3, 2, 0).

(c) specify which plane is acute angle bisector and which one is obtuse angle bisector.

STRAIGHT LINE

20. DEFINITION :

A straight line in space is characterised by the intersection of two planes which are not parallel and, therefore, the equation of a straight line is present as a solution of the system constituted by the equations of the two planes : $a_1 x + b_1 y + c_1 z + d_1 = 0$; $a_2 x + b_2 y + c_2 z + d_2 = 0$

This form is also known as **unsymmetrical form**.

Some particular straight lines :

	Straight lines	Equation
(i)	Through the origin	$y = mx, z = nx$
(ii)	x-axis	$y = 0, z = 0$ or $\frac{x}{1} = \frac{y}{0} = \frac{z}{0}$
(iii)	y-axis	$x = 0, z = 0$ or $\frac{x}{0} = \frac{y}{1} = \frac{z}{0}$
(iv)	z-axis	$x = 0, y = 0$ or $\frac{x}{0} = \frac{y}{0} = \frac{z}{1}$
(v)	parallel to x-axis	$y = p, z = q$
(vi)	parallel to y-axis	$x = h, z = q$
(vii)	parallel to z-axis	$x = h, y = p$

21. EQUATION OF A STRAIGHT LINE IN SYMMETRICAL FORM :

(a) One point form : Let $A(x_1, y_1, z_1)$ be a given point on the straight line and ℓ, m, n be the d.c's of the line, then its equation is

$$\frac{x - x_1}{\ell} = \frac{y - y_1}{m} = \frac{z - z_1}{n} = r \quad \text{(say)}$$

It should be noted that $P(x_1 + \ell r, y_1 + mr, z_1 + nr)$ is a general point on this line at a distance r from the point $A(x_1, y_1, z_1)$ i.e. AP = r. One should note that for AP = r; ℓ, m, n must be d.c.'s not d.r.'s. If a, b, c are direction ratios of the line, then equation of the line is

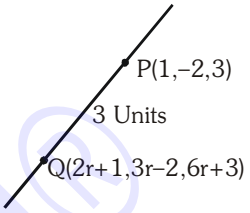
$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c} = r \quad \text{but here AP} \neq r$$

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(b) Equation of the line through two points $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$ is

$$\frac{x-x_1}{x_2-x_1}=\frac{y-y_1}{y_2-y_1}=\frac{z-z_1}{z_2-z_1}$$

Illustration 18 : Find the co-ordinates of those points on the line $\frac{x-1}{2}=\frac{y+2}{3}=\frac{z-3}{6}$ which is at a distance of 3 units from point (1,–2, 3).



Solution : Here, $\frac{x-1}{2}=\frac{y+2}{3}=\frac{z-3}{6}$ .....(i)

is the given straight line

Let, P = (1,–2,3) on the straight line

Here direction ratios of line (i) are (2,3,6)

$\therefore$ Direction cosines of line (i) are : $\frac{2}{7},\frac{3}{7},\frac{6}{7}$

$\Rightarrow$ Equations of line (i) can be written as

$$\frac{x-1}{2/7}=\frac{y+2}{3/7}=\frac{z-3}{6/7} \qquad \text{.....(ii)}$$

Co-ordinates of any point on the line (ii) can be taken as $\left(\frac{2}{7}r+1,\frac{3}{7}r-2,\frac{6}{7}r+3\right)$

Let, $Q\left(\frac{2}{7}r+1,\frac{3}{7}r-2,\frac{6}{7}r+3\right)$

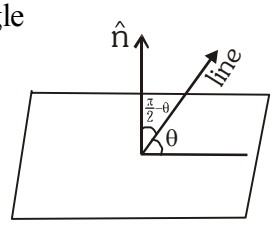
Given $|\vec{r}| = 3, \therefore r = \pm 3$

Putting the value of r, we have $Q\left(\frac{13}{7},-\frac{5}{7},\frac{39}{7}\right)$ or $Q=\left(\frac{1}{7},-\frac{23}{7},\frac{3}{7}\right)$ **Ans.**

---

22. ANGLE BETWEEN A LINE AND A PLANE :

Let equations of the line and plane be $\frac{x-x_1}{\ell}=\frac{y-y_1}{m}=\frac{z-z_1}{n}$ and $ax + by + cz + d = 0$ respectively and θ be the angle which line makes with the plane. Then $(\pi/2 - \theta)$ is the angle between the line and the normal to the plane.



So, $\sin\theta=\frac{a\ell+bm+cn}{\sqrt{(a^2+b^2+c^2)}\sqrt{(\ell^2+m^2+n^2)}}$

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Line is parallel to plane if $\theta = 0$ i.e. if $a\ell + bm + cn = 0$.

Line is perpendicular to the plane if line is parallel to the normal of the plane i.e. if $\frac{a}{\ell} = \frac{b}{m} = \frac{c}{n}$.

Illustration 19 : Find the angle between the line $\frac{x-2}{3} = \frac{y+1}{-1} = \frac{z-3}{-2}$ and the plane $3x + 4y + z + 5 = 0$.

Solution : The given line is $\frac{x-2}{3} = \frac{y+1}{-1} = \frac{z-3}{-2}$ ..... (i)

and the given plane is $3x + 4y + z + 5 = 0$ ..... (ii)

If the line (i) makes angle θ with the plane (ii), then the line (i) will make angle $(90° - \theta)$ with the normal to the plane (i). Now direction-ratios of line (i) are $3, -1, -2$ and direction-ratios of normal to plane (ii) are $3, 4, 1$

$$\therefore \quad \cos(90° - \theta) = \frac{(3)(3) + (-1)(4) + (-2)(1)}{\sqrt{9+1+4}\sqrt{9+16+1}} \quad \Rightarrow \quad \sin\theta = \frac{9-4-2}{\sqrt{14}\sqrt{26}} = \frac{3}{\sqrt{14}\sqrt{26}}$$

Hence $\theta = \sin^{-1}\left(\frac{3}{\sqrt{14}\sqrt{26}}\right)$ **Ans.**

Do yourself - 8 :

1. Find the equation of the line passing through the point (4, 2, 3) and having direction ratios 1, –1, 2

2. Find the symmetrical form of the line $x - y + 2z = 5$, $3x + y + z = 6$.

3. Find the angle between the plane $3x + 4y + 5 = 0$ and the line $\frac{x-1}{2} = \frac{y-2}{0} = \frac{z-1}{1}$.

4. Prove that the line $\frac{x-3}{2} = \frac{y-4}{3} = \frac{z-5}{4}$ is parallel to the plane $4x + 4y - 5z + 2 = 0$.

5. The line $\frac{x-x_1}{0} = \frac{y-y_1}{1} = \frac{z-z_1}{2}$ is

(A) parallel to x-axis (B) perpendicular to x-axis

(C) perpendicular to YOZ plane (D) parallel to y-axis

6. Find the equations of the straight line passing through the point (1,2,3) to intersect the straight line $x + 1 = 2(y - 2) = z + 4$ and parallel to the plane $x + 5y + 4z = 0$.

7. A tetrahedron has vertices P(1, 2, 1), Q(2, 1, 3), R(–1,1,2) and O(0, 0, 0). The angle between the faces OPQ and PQR is :

(A) $\cos^{-1}\left(\frac{9}{35}\right)$ (B) $\cos^{-1}\left(\frac{19}{35}\right)$ (C) $\cos^{-1}\left(\frac{17}{31}\right)$ (D) $\cos^{-1}\left(\frac{7}{31}\right)$

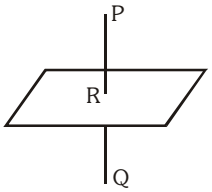
23. CONDITION THAT A LINE LIES ON THE GIVEN PLANE :

The line $\frac{x-x_1}{\ell} = \frac{y-y_1}{m} = \frac{z-z_1}{n}$ will lie on the plane $Ax + By + Cz + D = 0$ if

(a) $A\ell + Bm + Cn = 0$ and **(b)** $Ax_1 + By_1 + Cz_1 + D = 0$

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24. IMAGE OF A POINT IN THE PLANE :



In order to find the image of a point $P(x_1, y_1, z_1)$ in a plane $ax + by + cz + d = 0$, assume it as a mirror. Let $Q(x_2, y_2, z_2)$ be the image of the point $P(x_1, y_1, z_1)$ in the plane, then

(a) Line PQ is perpendicular to the plane. Hence equation of PQ is $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c} = r$

(b) Hence, Q satisfies the equation of line then $\frac{x_2-x_1}{a} = \frac{y_2-y_1}{b} = \frac{z_2-z_1}{c} = r$. The plane passes through the middle point of line PQ, therefore the middle point satisfies the equation of the plane i.e. $a\left(\frac{x_2+x_1}{2}\right) + b\left(\frac{y_2+y_1}{2}\right) + c\left(\frac{z_2+z_1}{2}\right) + d = 0$. The co-ordinates of Q can be obtained by solving these equations.

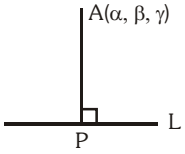
25. FOOT, LENGTH AND EQUATION OF PERPENDICULAR FROM A POINT TO A LINE :

Let equation of the line be $\frac{x-x_1}{\ell} = \frac{y-y_1}{m} = \frac{z-z_1}{n} = r$ .....(i)

and A (α, β, γ) be the point. Any point on the line (i) is $P(\ell r + x_1, mr + y_1, nr + z_1)$.....(ii)

If it is the foot of the perpendicular, from A on the line, then AP is $\perp$ to the line, so

$$\ell(\ell r + x_1 - \alpha) + m(mr + y_1 - \beta) + n(nr + z_1 - \gamma) = 0$$



i.e. $r = (\alpha - x_1)\ell + (\beta - y_1)m + (\gamma - z_1)n$

since $\ell^2 + m^2 + n^2 = 1$

Putting this value of r in (ii), we get the foot of perpendicular from point A to the line.

Length : Since foot of perpendicular P is known, length of perpendicular,

$$AP = \sqrt{[(\ell r + x_1 - \alpha)^2 + (mr + y_1 - \beta)^2 + (nr + z_1 - \gamma)^2]}$$

Equation of perpendicular is given by

$$\frac{x-\alpha}{\ell r + x_1 - \alpha} = \frac{y-\beta}{mr + y_1 - \beta} = \frac{z-\gamma}{nr + z_1 - \gamma}$$

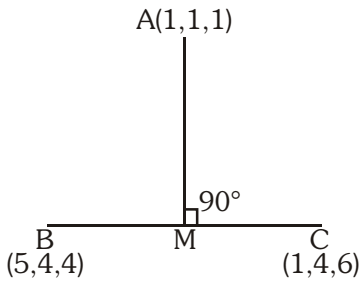
Illustration 20 : Find the co-ordinates of the foot of the perpendicular from (1, 1, 1) on the line joining (5, 4, 4) and (1, 4, 6).

Solution : Let A (1, 1, 1), B (5, 4, 4) and C (1, 4, 6) be the given points. Let M be the foot of the perpendicular from A on BC.

If M divides BC in the ratio λ : 1, then

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co-ordinates of M are $\left(\frac{\lambda+5}{\lambda+1},\frac{4\lambda+4}{\lambda+1},\frac{6\lambda+4}{\lambda+1}\right)$



Direction ratios of BC are 1 – 5, 4 – 4, 6 – 4
i.e. –4, 0, 2

D.R.'s of AM are $\frac{\lambda+5}{\lambda+1}-1,\frac{4\lambda+4}{\lambda+1}-1,\frac{6\lambda+4}{\lambda+1}-1$

$\Rightarrow \quad \frac{4}{\lambda+1},\frac{3\lambda+3}{\lambda+1},\frac{5\lambda+3}{\lambda+1} \quad \Rightarrow \quad 4,\ 3\lambda+3,\ 5\lambda+3$

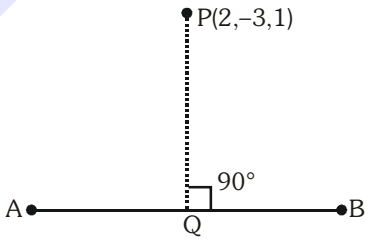
Since AM $\perp$ BC

$\therefore \quad 2(4)+0(3\lambda+3)-1(5\lambda+3)=0 \quad \Rightarrow 8-5\lambda-3=0 \quad \Rightarrow \quad \lambda=1$

Hence the co-ordinates of M are (3, 4, 5) **Ans.**

Illustration 21 : Find the length of perpendicular from P(2, – 3,1) to the line $\frac{x+1}{2}=\frac{y-3}{3}=\frac{z+2}{-1}$

Solution : Given line is $\frac{x+1}{2}=\frac{y-3}{3}=\frac{z+2}{-1}$ .....(i)



and P(2, – 3,1)

Co-ordinates of any point on (i) may be taken as
(2r–1,3r+3,–r–2)

Let Q = (2r –1, 3r + 3,–r – 2)

Direction ratio's of PQ are : (2r –3, 3r +6, –r –3)

Direction ratio's of AB are : (2, 3, –1)

Since, PQ $\perp$ AB

2 (2r – 3) +3 (3r + 6) – 1 (–r – 3) = 0

$\Rightarrow \quad r=-\frac{15}{14}$

$\therefore \quad Q=\left(-\frac{22}{7},-\frac{3}{14},-\frac{13}{14}\right)$

$$PQ^2=\left(2+\frac{22}{7}\right)^2+\left(-3+\frac{3}{14}\right)^2+\left(1+\frac{13}{14}\right)^2=\frac{531}{14}$$

$$PQ=\sqrt{\frac{531}{14}}\text{ units}$$ **Ans.**

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Do yourself - 9 :

1. Find the image of point P(1, 3, 2) in the plane $2x - y + z + 3 = 0$ as well as the foot of the perpendicular drawn from the point (1, 3, 2).

2. Find the distance of the point (1, –2, 3) from the plane $x - y + z = 5$ measured parallel to the line $\frac{x}{2} = \frac{y}{3} = \frac{z}{-6}$

3. Prove that $\frac{x+1}{-2} = \frac{y+2}{3} = \frac{z+5}{4}$ lies in the plane $x + 2y - z = 0$.

4. Feet of the perpendicular drawn from the point P(2,3,–5) on the axes of coordinates are A,B and C. Find the equation of the plane passing through their feet and the area of ΔABC.

5. If the lines $\frac{x-2}{1} = \frac{y-3}{1} = \frac{z-4}{-k}$ and $\frac{x-1}{k} = \frac{y-4}{2} = \frac{z-5}{1}$ are coplanar, then k can have :

(A) any value (B) exactly one value
(C) exactly two values (D) exactly three values.

6. If the distance of the point P(1,–2,1) from the plane $x + 2y - 2z = \alpha$, where $\alpha > 0$, is 5, then the foot of the perpendicular from P to the plane is-

(A) $\left(\frac{8}{3},\frac{4}{3},-\frac{7}{3}\right)$ (B) $\left(\frac{4}{3},-\frac{4}{3},\frac{1}{3}\right)$ (C) $\left(\frac{1}{3},\frac{2}{3},\frac{10}{3}\right)$ (D) $\left(\frac{2}{3},-\frac{1}{3},\frac{5}{2}\right)$

7. The point P is the intersection of the straight line joining the points Q(2,3,5) and R(1,–1,4) with the plane $5x - 4y - z = 1$. If S is the foot of the perpendicular drawn from the point T(2,1,4) to QR, then the length of the line segment PS is -

(A) $\frac{1}{\sqrt{2}}$ (B) $\sqrt{2}$ (C) 2 (D) $2\sqrt{2}$

8. If the straight lines $\frac{x-1}{2} = \frac{y+1}{k} = \frac{z}{2}$ and $\frac{x+1}{5} = \frac{y+1}{2} = \frac{z}{k}$ are coplanar, then the plane(s) containing these two lines is(are)

(A) $y + 2z = -1$ (B) $y + z = -1$ (C) $y - z = -1$ (D) $y - 2z = -1$

26. EQUATION OF PLANE CONTAINING TWO INTERSECTING LINES :

Let the two lines be

$$\frac{x-\alpha_1}{\ell_1} = \frac{y-\beta_1}{m_1} = \frac{z-\gamma_1}{n_1} \quad \text{..... (i)}$$

and

$$\frac{x-\alpha_2}{\ell_2} = \frac{y-\beta_2}{m_2} = \frac{z-\gamma_2}{n_2} \quad \text{..... (ii)}$$

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These lines will coplanar if $\begin{vmatrix} \alpha_2-\alpha_1 & \beta_2-\beta_1 & \gamma_2-\gamma_1 \\ \ell_1 & m_1 & n_1 \\ \ell_2 & m_2 & n_2 \end{vmatrix} = 0$ (It is condition for intersection of two lines)

the plane containing the two lines is $\begin{vmatrix} x-\alpha_1 & y-\beta_1 & z-\gamma_1 \\ \ell_1 & m_1 & n_1 \\ \ell_2 & m_2 & n_2 \end{vmatrix} = 0$

Illustration 22 : Find the equation of the plane containing the line $\frac{x-1}{3}=\frac{y+6}{4}=\frac{z+1}{2}$ and parallel to the line $\frac{x-4}{2}=\frac{y-1}{-3}=\frac{z+3}{5}$.

Solution : Any plane containing the line $\frac{x-1}{3}=\frac{y+6}{4}=\frac{z+1}{2}$ is

$a(x-1)+b(y+6)+c(z+1)=0$ ..... (i)

where, $3a+4b+2c=0$ ..... (ii)

Also, it is parallel to the second line and hence, its normal is perpendicular to this line

$\therefore$ $2a-3b+5c=0$ ..... (iii)

Solving (ii) & (iii) by cross multiplication, we get $\frac{a}{26}=\frac{b}{-11}=\frac{c}{-17}=k$

$\Rightarrow$ $a=26k,\ b=-11k\ \&\ c=-17k$

Putting these values in (i), we get $26k(x-1)-11k(y+6)-17k(z+1)=0$

$\Rightarrow$ $26x-11y-17z=109$, which is the required equation of the plane.

27. LINE OF GREATEST SLOPE :

Consider two planes G-plane and H-plane. H-plane is treated as a horizontal plane or reference plane. G-plane is a given plane. Let AB be the line of intersection of G-plane & H-plane. Line of greatest slope is a line which is contained by G-plane & perpendicular to line of intersection of G-plane & H-plane. Obviously, infinitely many such lines of greatest slopes are contained by G-plane. Generally an additional information is given in problem so that a unique line of greatest slope can be found out.

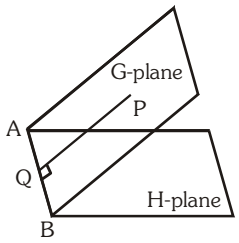


Illustration 23 : Assuming the plane $4x-3y+7z=0$ to be horizontal, find the equation of the line of greatest slope on the plane $2x+y-5z=0$, passing through the point (2,1,1).

Solution : The required line passing through the point P (2,1,1) in the plane $2x+y-5z=0$ and is having greatest slope, so it must be perpendicular to the line of intersection of the planes

$2x+y-5z=0$ .....(i)

and $4x-3y+7z=0$ .....(ii)

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Let the d.r'.s of the line of intersection of (i) and (ii) be a, b, c

$\Rightarrow$ $2a + b - 5c = 0$ and $4a - 3b + 7c = 0$

{as dr'.s of straight line (a, b,c) is perpendicular to d.r'.s of normal to both the planes}

$\Rightarrow$ $\frac{a}{4} = \frac{b}{17} = \frac{c}{5}$

Now let the direction ratio of required line be proportional to ℓ, m, n then its

equation be $\frac{x-2}{\ell} = \frac{y-1}{m} = \frac{z-1}{n}$

where $2\ell + m - 5n = 0$ and $4\ell + 17m + 5n = 0$

so, $\frac{\ell}{3} = \frac{m}{-1} = \frac{n}{1}$

Thus the required line is $\frac{x-2}{3} = \frac{y-1}{-1} = \frac{z-1}{1}$ **Ans.**

28. AREA OF TRIANGLE :

To find the area of a triangle in terms of its projections on the co-ordinates planes.

Let Δ_x, Δ_y, Δ_z be the projections of the plane area of the triangle on the planes yOz, zOx, xOy respectively.Let ℓ, m, n be the direction cosines of the normal to the plane of the triangle.

Then the angle between the plane of the triangle and yOz plane is the angle between the normal to the plane of the triangle and the x-axis.

$\therefore$ $\Delta_x = \Delta\ell$

Similarly $\Delta_y = \Delta m$; $\Delta_z = \Delta n$ $\Rightarrow$ $\Delta = \sqrt{\Delta_x^2 + \Delta_y^2 + \Delta_z^2}$

If $A(x_1, y_1, z_1)$, $B(x_2, y_2, z_2)$, $C(x_3, y_3, z_3)$ be the three vertices of the triangle then

$$\Delta_x = \frac{1}{2}\begin{vmatrix} y_1 & z_1 & 1 \\ y_2 & z_2 & 1 \\ y_3 & z_3 & 1 \end{vmatrix}, \Delta_y = \frac{1}{2}\begin{vmatrix} x_1 & z_1 & 1 \\ x_2 & z_2 & 1 \\ x_3 & z_3 & 1 \end{vmatrix}, \Delta_z = \frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

Do yourself - 10 :

1. Prove that the lines $\frac{x+1}{3} = \frac{y+3}{5} = \frac{z+5}{7}$ and $\frac{x-2}{1} = \frac{y-4}{3} = \frac{z-6}{5}$ are coplanar. Find their point of intersection.

2. Find the area of the triangle whose vertices are the points (1, 2, 3), (–2, 1, –4), (3, 4, –2).

3. The lines $\frac{x-2}{1} = \frac{y-3}{1} = \frac{z-4}{-k}$ and $\frac{x-1}{k} = \frac{y-4}{2} = \frac{z-5}{1}$ are coplanar if

(A) k = 0 or – 1 (B) k = 1 or – 1 (C) k = 0 or – 3 (D) k = 3 or – 3

4. Find the equations to the line of greatest slope through the point (7, 2, –1) in the plane $x-2y+3z=0$ assuming that the axes are so placed that the plane $2x+3y-4z=0$ is horizontal.

5. If A, B, C, D are any four points in space, then $|\overrightarrow{AB}\times\overrightarrow{CD}+\overrightarrow{BC}\times\overrightarrow{AD}+\overrightarrow{CA}\times\overrightarrow{BD}|$ equals -

(A) area of ΔABC (B) area of ΔBCD

(C) area of ΔCDA (D) 4(area of ΔABC)

6. Equation of the plane containing the straight line $\frac{x}{2}=\frac{y}{3}=\frac{z}{4}$ and perpendicular to the plane containing the straight lines $\frac{x}{3}=\frac{y}{4}=\frac{z}{2}$ and $\frac{x}{4}=\frac{y}{2}=\frac{z}{3}$ is

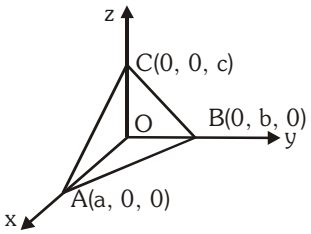
(A) $x + 2y - 2z = 0$ (B) $3x + 2y - 2z = 0$

(C) $x - 2y + z = 0$ (D) $5x + 2y - 4z = 0$

Miscellaneous Illustrations :

Illustration 24 : If a variable plane cuts the coordinate axes in A, B and C and is at a constant distance p from the origin, find the locus of the centre of the tetrahedron OABC.

Solution : Let $A \equiv (a, 0, 0)$, $B \equiv (0, b, 0)$ and $C \equiv (0, 0, c)$



$\therefore$ Equation of plane ABC is $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$

Now p = length of perpendicular from O to this plane

$$= \frac{1}{\sqrt{\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}}} \quad \text{or} \quad p^2 = \frac{1}{\left(\frac{1}{a}\right)^2+\left(\frac{1}{b}\right)^2+\left(\frac{1}{c}\right)^2} \quad ...(i)$$

Let $G(\alpha, \beta, \gamma)$ be the centre of the tetrahedron OABC, then

$$\alpha = \frac{a}{4}, \beta = \frac{b}{4}, \gamma = \frac{c}{4} \qquad \left[\because \alpha = \frac{a+0+0+0}{4} = \frac{a}{4}\right]$$

or, $a = 4\alpha, b = 4\beta, c = 4\gamma$

Putting these values of a, b, c in equation (i), we get

$$p^2 = \frac{16}{\left(\frac{1}{\alpha^2}+\frac{1}{\beta^2}+\frac{1}{\gamma^2}\right)} \quad \text{or} \quad \frac{1}{\alpha^2}+\frac{1}{\beta^2}+\frac{1}{\gamma^2}=\frac{16}{p^2}$$

$\therefore$ locus of (α, β, γ) is $x^{-2} + y^{-2} + z^{-2} = 16\, p^{-2}$ **Ans.**

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Illustration 25 : Through a point P(h, k, ℓ) a plane is drawn at right angles to OP to meet the coordinate axes in A, B and C. If OP = p, show that the area of ΔABC is $\frac{p^5}{2|hk\ell|}$.

Solution : $OP = \sqrt{h^2+k^2+\ell^2} = p$

Direction cosines of OP are $\frac{h}{\sqrt{h^2+k^2+\ell^2}}, \frac{k}{\sqrt{h^2+k^2+\ell^2}}, \frac{\ell}{\sqrt{h^2+k^2+\ell^2}}$

Since OP is normal to the plane, therefore, equation of the plane will be,

$$\frac{h}{\sqrt{h^2+k^2+\ell^2}}x+\frac{k}{\sqrt{h^2+k^2+\ell^2}}y+\frac{\ell}{\sqrt{h^2+k^2+\ell^2}}z=\sqrt{h^2+k^2+\ell^2}$$

or, $hx + ky + \ell z = h^2 + k^2 + \ell^2 = p^2$

$$\therefore \quad A \equiv \left(\frac{p^2}{h}, 0, 0\right), B \equiv \left(0, \frac{p^2}{k}, 0\right), C \equiv \left(0, 0, \frac{p^2}{\ell}\right)$$

Now area of ΔABC, $\Delta^2 = A_{xy}^2 + A_{yz}^2 + A_{zx}^2$

Now A_{xy} = area of projection of ΔABC on xy-plane = area of ΔAOB

$$= \text{Mod of } \frac{1}{2}\begin{vmatrix} \frac{p^2}{h} & 0 & 1 \\ 0 & \frac{p^2}{k} & 1 \\ 0 & 0 & 1 \end{vmatrix} = \frac{1}{2}\frac{p^4}{|hk|}$$

Similarly, $A_{yz} = \frac{1}{2}\frac{p^4}{|k\ell|}$ and $A_{zx} = \frac{1}{2}\frac{p^4}{|\ell h|}$

$$\therefore \quad \Delta^2 = \frac{1}{4}\frac{p^8}{h^2k^2}+\frac{1}{4}\frac{p^8}{k^2\ell^2}+\frac{1}{4}\frac{p^8}{h^2\ell^2} = \frac{p^{10}}{4h^2k^2\ell^2}$$

or $\Delta = \frac{p^5}{2|hk\ell|}$ **Ans.**

Illustration 26 : Find the locus of a point, the sum of squares of whose distances from the planes : $x - z = 0$, $x - 2y + z = 0$ and $x + y + z = 0$ is 36

Solution : Given planes are $x - z = 0$, $x - 2y + z = 0$ and, $x + y + z = 0$

Let the point whose locus is required be $P(\alpha,\beta,\gamma)$. According to question

$$\frac{|\alpha-\gamma|^2}{2}+\frac{|\alpha-2\beta+\gamma|^2}{6}+\frac{|\alpha+\beta+\gamma|^2}{3}=36$$

or $3(\alpha^2+\gamma^2-2\alpha\gamma)+\alpha^2+4\beta^2+\gamma^2-4\alpha\beta-4\beta\gamma+2\alpha\gamma+2(\alpha^2+\beta^2+\gamma^2+2\alpha\beta+2\beta\gamma+2\alpha\gamma) = 36\times 6$

or $6\alpha^2 + 6\beta^2 + 6\gamma^2 = 36 \times 6$

or $\alpha^2 + \beta^2 + \gamma^2 = 36$

Hence, the required equation of locus is $x^2 + y^2 + z^2 = 36$ **Ans.**

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Illustration 27 : Direction ratios of normal to the plane which passes through the point (1, 0, 0) and (0, 1, 0) which makes angle $\pi/4$ with $x + y = 3$ are -

(A) 1, 1, 2 (B) $\sqrt{2}$, 1, 1 (C) 1, $\sqrt{2}$, 1 (D) 1, 1, $\sqrt{2}$

Solution : The plane by intercept form is $\frac{x}{1}+\frac{y}{1}+\frac{z}{c}=1$

direction ratios of normal of this plane are $1, 1, \frac{1}{c}$ and that of given plane are 1, 1, 0.

$$\therefore \quad \cos\frac{\pi}{4}=\frac{1\,.\,1+1\,.\,1+0\cdot\frac{1}{c}}{\sqrt{1+1+\frac{1}{c^2}}\,\sqrt{1+1+0}}$$

$$\Rightarrow \quad \frac{1}{\sqrt{2}}=\frac{2}{\sqrt{2+\frac{1}{c^2}}\,\sqrt{2}} \Rightarrow 2+\frac{1}{c^2}=4 \Rightarrow \quad c=\pm\frac{1}{\sqrt{2}}$$

$\therefore$ d.r.'s are 1, 1, $\sqrt{2}$ **Ans. (D)**

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